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# Wilful ignorance at Waterloo: public housing quality and political stigma in Sydney's largest estate renewal

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**ABSTRACT**

Sydney's Waterloo estate is one of Australia's largest and most concentrated areas of public housing. There has been an almost two-decade-long effort to renew the estate, with successive governments persistently making the case that the demolition and mixed-tenure redevelopment must happen. In the process, the quality of the housing on the estate has been denigrated and misrepresented. This paper combines a detailed examination of the architectural qualities of the Waterloo estate with a critical analysis of historical and contemporary policy settings and political decisions driving the redevelopment project. It identifies the phases of development at Waterloo since the first buildings were constructed in 1948, describing the key characteristics of the dwellings that were part of each phase and situating them within in the wider story of public housing in NSW. It then evaluates the qualities identified in the buildings on the estate against the government's own guidelines for new apartment development. This evaluation reveals the extent to which the state government has avoided a good-faith assessment of the architectural quality of the estate. The paper concludes by reviewing planning and policy objectives and highlighting that tenant interests and housing quality have been obscured through the stigmatzation of the estate.

**KEYWORDS**

Public housing; Estate renewal; Architecture; Design guidelines; Sydney

## Introduction

The history of public housing in Australia is being rewritten by redevelopment. Estate renewal, which has come to mean demolition of public housing estates and the construction of mixed-tenure residential developments in their place, has gathered pace over the past two decades, mirroring an international trend.[1] State governments, particularly in Victoria and New South Wales (NSW), have driven these transformations on broadacre outer suburban estates as well as in more complex, high-density inner-city settings.[2] In both cases, policy documents and politicians' pronouncements describe existing dwellings and neighbourhoods as outdated and unfit for twenty-first-century tenants. The estates and their buildings are often cast as failed experiments that are responsible

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[1]Goetz, *New Deal Ruins*; Watt, *Estate Regeneration and its Discontents*; Watt and Smets (eds.), *Social Housing and Urban Renewal*.

[2]Arthurson, *Social Mix and the City*; Darcy, "From High-Rise Projects to Suburban Estates"; Wynne et al., "The Marketisation of Social Housing".

This article has been corrected with minor changes. These changes do not impact the academic content of the article.

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for social dysfunction. Blame is variously attributed to the apparent hubris of mid-century architects and planners, who are accused of embracing and promoting untried and ill-conceived models of development; technical failures and compromises, due to austere financial conditions imposed on their design and construction; and functional obsolescence.[3] While these explanations are often evoked in overlapping ways that lack a coherent accompanying analysis, the calculation that sits behind them is clear. The political leaders and policy advisers who promote this vision of estate renewal intend their audiences to be left with the very clear impression that twentieth-century public housing falls far short of contemporary standards and norms. They rarely shoulder responsibility for any of these failings. But the estates that have been identified as needing renewal are condemned as simply 'unfit for purpose'.

Such is the case with the redevelopment of the Waterloo estate: the largest estate renewal proposal to date, and one of the largest that could ever be undertaken in Australia. The estate comprises approximately 2000 dwellings across a range of high-, mid-, and low-rise apartment buildings on 18 hectares in Sydney's inner-south. In December 2015, the NSW Government announced that it would be included as part of the programme formerly known as 'Communities Plus'; the plan was to demolish the existing buildings and construct seven to eight thousand new apartments in their place, with approximately 30% of these social housing and 70% private housing. The effect of the project would have been to increase slightly the number of social housing units, although the new units would generally have less bedrooms and be able to house less people than those they replaced. The transfer of a portion of the estate into private ownership and sale of the private market units would fund the project as a whole. In August 2023, following a change of government in March 2023, the plan for the project's first phase – the redevelopment of 749 units in the predominantly low-rise Waterloo South – was revised to 30% social housing, 50% market-rate private housing, and 20% affordable housing (targeted and moderate-income earners, and leased for either a maximum of 30% of gross income or at a 20% discount to market rate).

In the face of a strong campaign by residents to stop or slow the redevelopment plans, the state government – first under the Liberal-National Coalition, and then under Labor since March 2023 – has highlighted the failings of the estate.[4] Former Minister for Social Housing Brad Hazzard initially declared that it was 'terrible'[5]; the Labor government's Minister for Housing an Homelessness, Rose Jackson, labelled it 'not fit for purpose'[6] and local Member of Parliament Ron Hoenig described it as 'well past the renewal date', with residents living in 'squalor'.[7] But are the units and the living environment on this estate really that bad? Is wholesale demolition for complete reconstruction the best option available to government? And does this approach properly consider those most affected, the existing residents? While resident groups have mounted a vigorous campaign against the project, most of the public debate has focused on how many new units might be acceptable on a redeveloped site. Other ways of conceiving of the estate – repairing, expanding, and refurbishing existing buildings over time, ensuring ongoing security for residents on site – appear not to have not been given serious consideration. One might assume then, given the proposals to date, that all the dwellings are entirely beyond redemption, unsalvageable. But the estate, which took more than a quarter of century to develop, is made up of several very different types of housing, at very different scales, with different plans, and employing completely different construction

[3]Sisson, "Public Housing and Territorial Stigma".
[4]Chaterjee et al., "Documentary and Resistance".
[5]Hazzard, "AHURI Conference Opening Address".
[6]McKeith "Public Housing Residents Protest".
[7]Wang, "Waterloo Estate Redevelopment".




**Figure 1.** 1975 aerial photo of the estate showing state of development with Matavai and Turanga towers under construction.

materials and methods. The government's all-encompassing appraisal, that they are no longer fit for purpose, throws a veil over the diversity and specificity of what is actually there. It asks us to take as read the idea that the whole estate is truly obsolete, that high-rise towers and slab blocks, as well as low-rise walk-ups and everything in between must go (Figure 1).

The purpose of this paper is to evaluate whether the demolition-redevelopment approach favoured by recent governments is ineluctable. The paper combines a detailed examination of the architectural qualities of the Waterloo estate with a critical analysis of historical and contemporary policies and political decisions driving the project. It provides a brief background that delineates the early history of the public housing programme in inner Sydney in the postwar decades before describing and analysing the qualities of each of the main phases of development of the Waterloo estate between the late 1940s and early 1980s, carefully unpacking the different qualities and characteristics of the buildings and their units. This provides basic information about the amenity of the dwellings – daylighting, natural ventilation, privacy, and so on. The paper then looks at how the qualities of the Waterloo estate's housing align or not with wider ideas about healthy environments and the other factors that inform the NSW Government's own housing quality standards for new apartments. The paper, furthermore, evaluates the extent to which the existing Waterloo units meet those standards.

This paper, however, is not a condition assessment nor an attempt to document exactly what residents want from their dwellings. It is an evaluation of the basic architectural qualities of the

buildings, the resultant amenity delivered and then an assessment of those aspects against the government's evidence-based design guidelines. Though the buildings and the units are of their time and somewhat mixed in quality, the results of this systematic evaluation raises questions about the underlying rationale for the state government's estate renewal plan as it evolved across the past decade. In particular, the paper is interested in whether the redevelopment plans – officially referred to as 'renewal' – gives due consideration to the needs of the tenants themselves and the possibility of more judicious redevelopment over time. And it asks why the government relies so heavily on stigmatizing the place and its inhabitants and yet seems to have done little in the way of systematic evaluation themselves.

## The emergence of public housing in NSW

A hundred years ago, Sydney faced a significant housing affordability and supply problem. The economic boom of the 1920s built prosperity for some, but many missed out, and housing for workers was often inadequate or simply unavailable.[8] The state government intervened to provide worker housing through its Housing Board (1910–1924), initially in a limited way in the inner-urban areas of Millers Point and The Rocks, and later more ambitiously at suburban Daceyville and other suburban and regional locations. However, prominent economic and political forces in NSW fiercely opposed this state support for housing and consequently its impact was quite limited in scope.[9] Under this scheme, the state government completed a total of 834 dwellings in greater Sydney for a population that passed one million during the 1920s.[10] Sydney Municipal Council also invested resources in workers' housing in this period (1912–1927), something entirely new to them. Their efforts reflected a clear vision, but were similarly limited in scope, providing just 165 dwellings for a population of more than 50,000 (Figure 2).[11]

In the first quarter of the twentieth century, planners and sanitation experts labelled many inner urban locations with strong concentrations of housing constructed in the period between 1850s and the 1890s as 'slums', and they were slated for resumption. But with limited provision of new state-provided housing, most such areas were redeveloped for profitable commercial or industrial uses, or alternatively just remained as they were with the condition of dwellings worsening. Private construction activity all but ceased during the Great Depression of the 1930s and Sydney faced a drastic housing shortage for its growing population.[12] In 1934, the State set up the Homes for the Unemployed Trust to build small cottages for subsidized sale and to supply cheap materials to enable people to build their own house in suburban areas. But the impact of the Unemployed Trust was also limited.[13]

This series of schemes was modest in terms of serving need but did create momentum for a more substantial public provision of housing. By the middle of the decade, a renewed slum-reform movement pushed for 'abolishing eye-sores' and providing better quality homes intended to create employment and improve the health of workers. The removal of terrace housing was a key feature of the initiative as this form of accommodation was widely viewed as 'ugly and monotonous', lacking in adequate services and subject to overcrowding.[14] It was the stigmatized housing of its day,

[8]Simpson, "Sydney Housing 1900 to 1940".
[9]Zanardo, "Shaping Affordable Housing," 179–234.
[10]Zanardo, "Shaping Affordable Housing", 249.
[11]Zanardo, "Shaping Affordable Housing", 264–5.
[12]Pettigrew, *Homes for the People*, 34.
[13]Zanardo, "Shaping Affordable Housing," 268–71.
[14]Spearritt, *Sydney's Century*, 70–1.



**Figure 2.** The Strickland Building is the first and largest example of workers' housing built by the Sydney Municipal Council, 1914.

and in response to pressure the State established the Housing Conditions Investigation Committee in 1936 to enquire into the issue and make recommendations. It concluded that urgent action was required, that the private sector could not or would not provide the quality of housing required without compromising its profits, and therefore that the Government needed to take on this responsibility. On the advice of the Committee, the State established the Housing Improvement Board in 1937 to undertake necessary slum clearance and construct housing schemes.[15] The Board investigated several slum areas (including Waterloo), but only one project was initiated.[16] Located in Erskineville and completed in 1938, the estate ultimately contained just 56 dwellings (Figure 3).[17]

The government halted further projects and funding at the outbreak of World War II in 1939. However, a change in government in 1941 brought with it an increased focus on addressing the questions of slum-clearance and housing. The new McKell Labor Government proposed a single new authority be established to 'co-ordinate and supervise housing throughout the State', and develop a 'long term programme to give the people homes in which they can live in dignity'.[18] In November 1941, the state government passed a new Housing Act establishing the Housing Commission of New South Wales. The Housing Commission assumed the powers and responsibilities

[15]Zanardo, "Shaping Affordable Housing," 277–8.
[16]Volke, "The Politics of State Rental Housing," 74.
[17]Zanardo, "Erskinville Rehousing Scheme," 78–9.
[18]"Australian Labor Party: Policy Speech," 4.



**Figure 3.** The Erskineville Re-Housing Scheme was developed by the Housing Improvement Board and completed in 1938.

of the previous Housing Board, Homes for the Unemployed Trust and the Housing Improvement Board and was charged with delivering 'adequate housing accommodation' across the State.[19]

The very first projects of the NSW Housing Commission were modest as material shortages and restrictions during and after World War II in Australia limited the state's ability to construct new buildings. Houses were built in country areas for those employed in the war effort, and soon after, temporary dwellings were erected in more urban areas for those in greatest need. However, at the end of the War, the Commonwealth lent money to the states under the first Commonwealth State Housing Agreement (CHSA) in 1945, allowing the Housing Commission to commence their activities in earnest.[20] Their initial projects were either new estates called 'neighbourhood developments' comprised of scores of small free-standing dwellings on large lots in greenfield areas, or smaller infill projects on resumed (or already available) sites in urban areas, often including modestly-scaled apartment buildings. Designs for standard two and three-storey apartment blocks were developed by the Housing Commission by in-house architects as early as 1946.

## (Re)making Waterloo

The first redevelopment at what is now known as the Waterloo Estate occurred in 1948 with four three-storey apartment buildings occupying a small site at the northern end of the urban block bounded by Raglan, George, Wellington and Cooper Streets, on land previously partially occupied by air-raid trenches. The buildings were arranged perpendicular to the street frontages with a

[19]Housing Act 1941 (NSW).
[20]Troy, *Accommodating Australians*, 95–100.

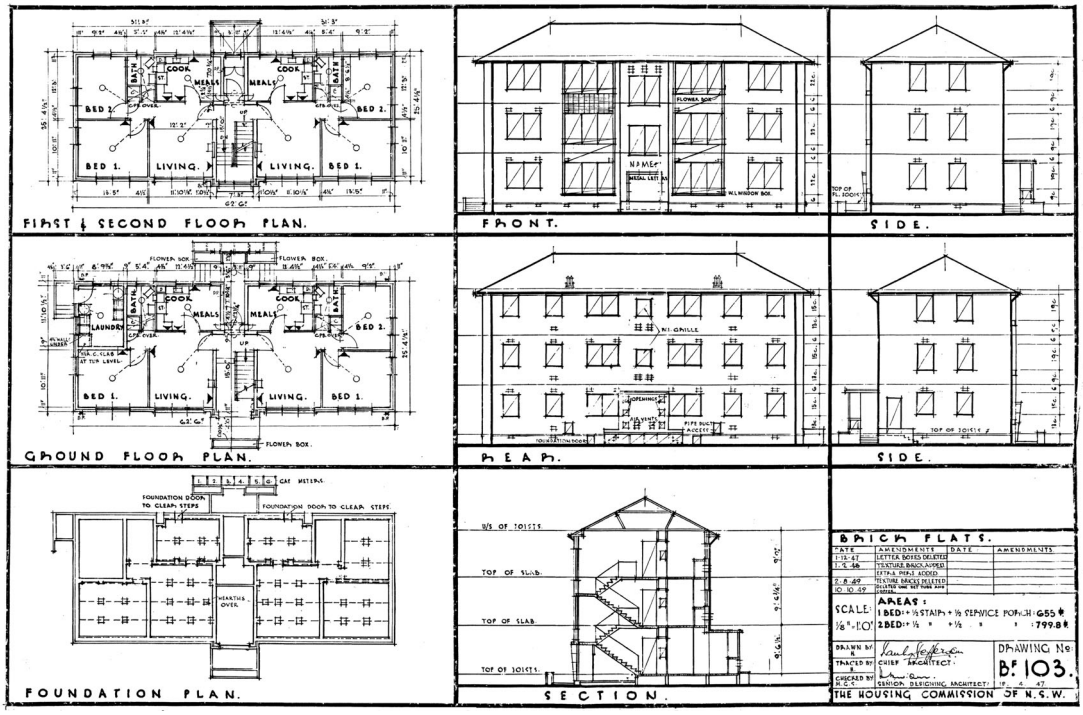



**Figure 4.** Standard brick flats building type BF103 with two two-bedroom units per floor.

landscaped setback and together formed a shared courtyard at the centre-rear of the site. Each small brick apartment building had a simple rectangular building footprint, a pitched, tile roof, and contained six two-bedroom units, two per floor, mirrored across a communal stair. Units were relatively small by today's standards but had highly efficient layouts with well-proportioned rooms in a pin-wheeled configuration around a minimal corridor, kitchen with sit-in meal room, and modern bathroom facilities. A very thin building section, high ceilings and windows to three sides of every unit provided excellent daylight to all areas of the plan and easy cross-ventilation. Few shared walls meant that there was little acoustic or impact noise transfer laterally (Figure 4).

The scale of urban redevelopment projects grew quickly as the state government made a concerted push to increase the volume of housing under construction. External architects worked with the Housing Commission to develop bespoke designs for site-specific circumstances on larger sites that had been acquired or assembled before World War II. In 1954, the Housing Commission completed Greenway in Kirribilli on land left over from the construction of the Sydney Harbour Bridge. Designed by D.T. Morrow & Gordon Architects, it contained 309 dwellings, making it the largest residential development in Australia at the time. John Northcott Place in Surry Hills was completed in 1963, this time on land originally assembled by the City of Sydney for housing. It was designed by Samuel Lipson and Kaad Architects and contained 409 dwellings again claiming the title of largest multi-storey public housing development in Australia (Figure 5).[21]

By the early 1960s, several entire urban street blocks in Waterloo had been acquired and cleared, rear lanes removed, and subdivisions amalgamated and prepared for redevelopment. Apartment

[21]Butler-Bowdon and Pickett, *Homes in the Sky*, 134–9; Glendinning, *Mass Housing*, 136–7.



**Figure 5.** The dramatic increase in scale of public housing in Sydney is illustrated here by the development of John Northcott Place built by the NSW Housing Commission and completed in 1963. It is surrounded by earlier walk-up public housing flats from 1953.

building designs were along the lines of the small original project in 1948; however, updated standard building plans were being utilized by this time (Figure 6). Rectangular buildings contained collections of either exclusively one-bedroom units, two-bedroom units or three-bedroom units, and were organized two per floor either side of a stair over three storeys. Newer cruciform plan types contained a combination of unit sizes but were collections of four per floor arranged around a central stairwell. Internally, these unit plans had many of the qualities and characteristics of the earlier ones, however, a couple of small differences from the 1948 building design are apparent, including meal rooms relocated to be part of an open-plan living room, the provision of a small balcony to each unit, and the lowering of ceilings.

While they appeared satisfied with this apartment building type, the Housing Commission introduced a significant change at the urban scale in this phase of development. With the availability of entire street blocks the Commission broke with the traditional pattern of aligning housing to the street frontage. They opted instead for alternative angled layouts which no longer strongly defined the street edges with built form. This arrangement instead provided increased and irregular landscaped setbacks and appear almost as if planned from the birds-eye view (Figures 7 and 8). Internally to the block, behind the building line, the space between the buildings became less

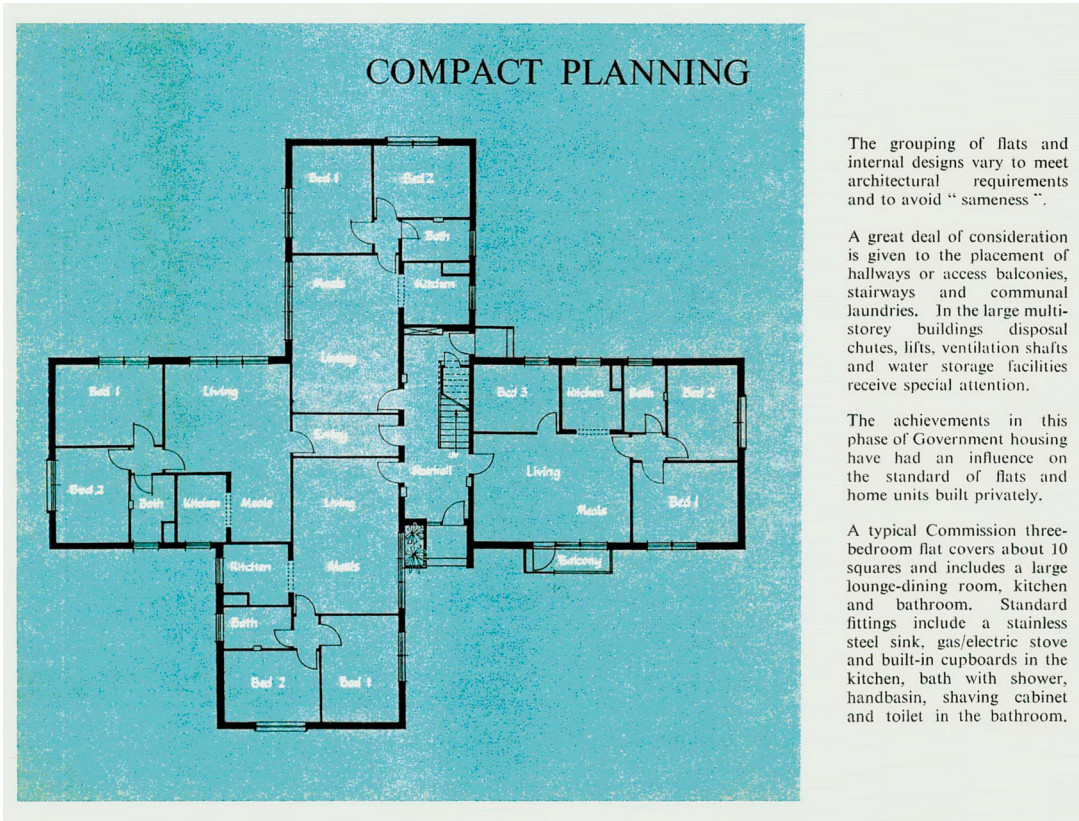



**Figure 6.** This typical plan for two- and three-bedroom units in housing commission flats highlights how dwellings could be effectively daylit and cross-ventilated.

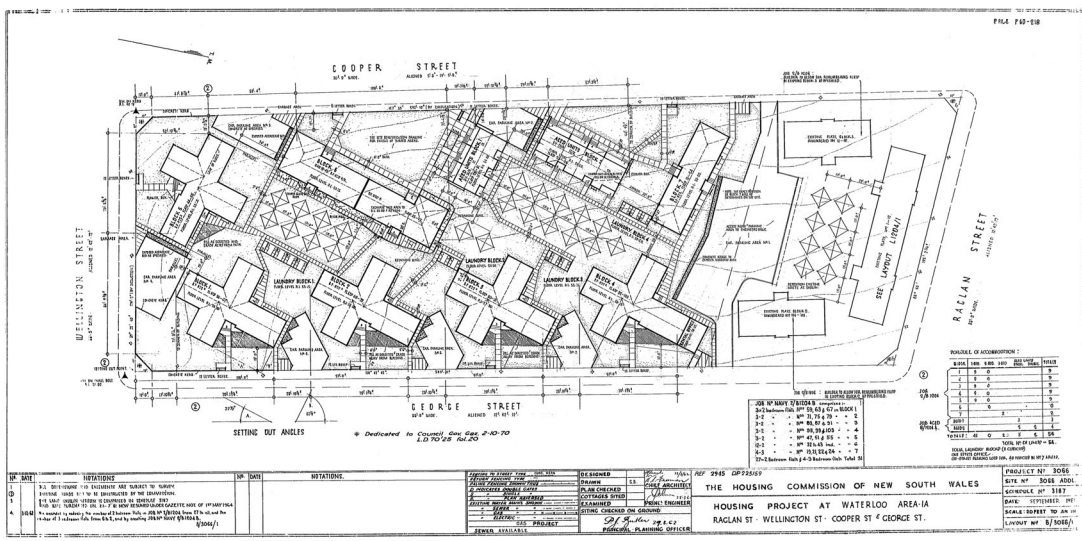



**Figure 7.** Standard cruciform and rectangular 3-storey building types set at angles to the street with 1948 rectangular layout to north (right) of the block.

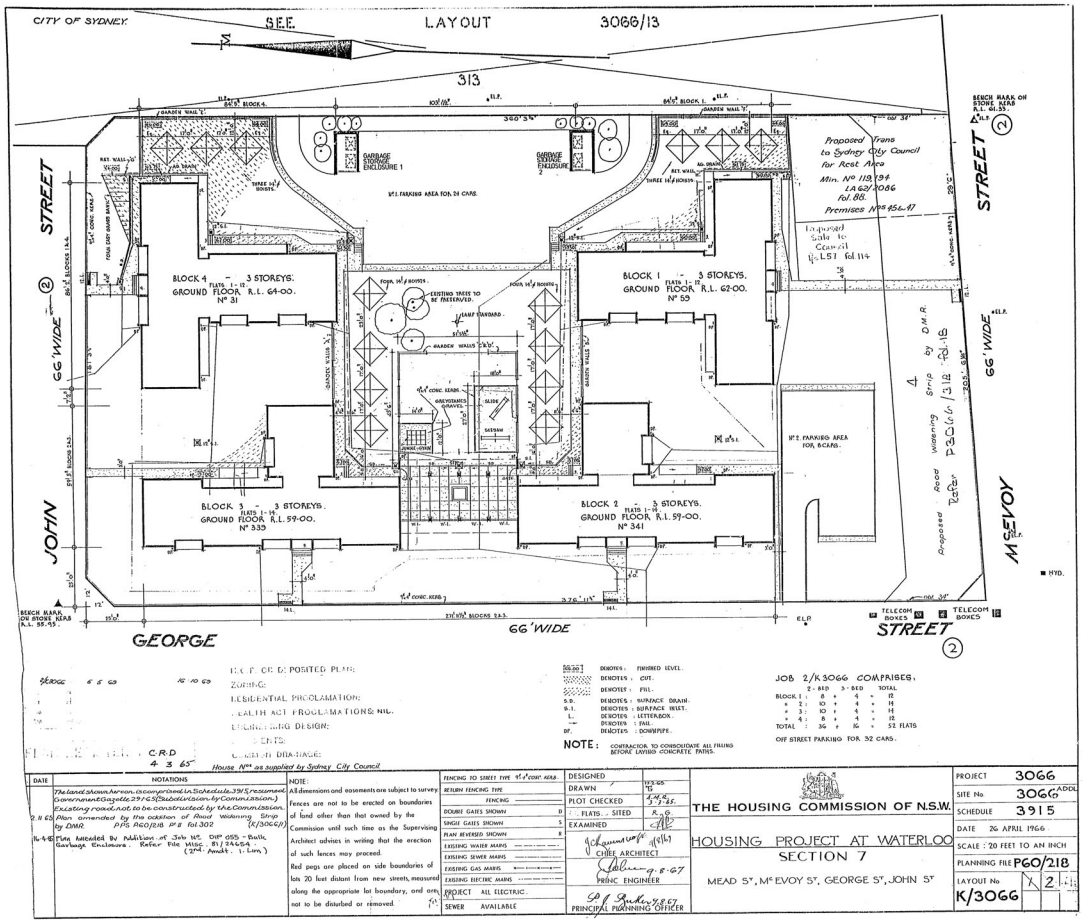



**Figure 8.** Block layout for Mount Lachlan showing alignment to street edge.

physically contained or courtyard-like, and more free-flowing. An advantage of the layout appears to be that windows do not directly face opposing buildings and outlook from units is extended further into landscaped spaces. A secondary consequence is that the building entries are not perpendicular to the footpath and are further from the street diminishing the sense of address and sense of 'front' and 'back', with buildings being perceived 'in the round'.

In 1968, a project called Mount Lachlan, which includes 52 dwellings, was designed for the southern edge of the Waterloo Estate in the block bounded by John, Mead, McEvoy and George Streets. Comprised of four larger 'T-shaped' three-storey buildings, each with four or five units per floor, the project was a return to the earlier 1948 approach of laying out buildings aligned to the street edge. The architectural language of Mount Lachlan is distinguished by a more modern sensibility with a flat roof, blade wall expression around balconies, and features a change in brick tone for spandrels between windows which helps to break up larger areas of wall plane reducing the apparent scale of the building. Another difference is the introduction of short, externalized corridors, or ambulatories, to reach units not immediately adjacent to the staircase. The dwellings themselves, as with the earlier buildings, are compact in plan, with windows on two or three sides

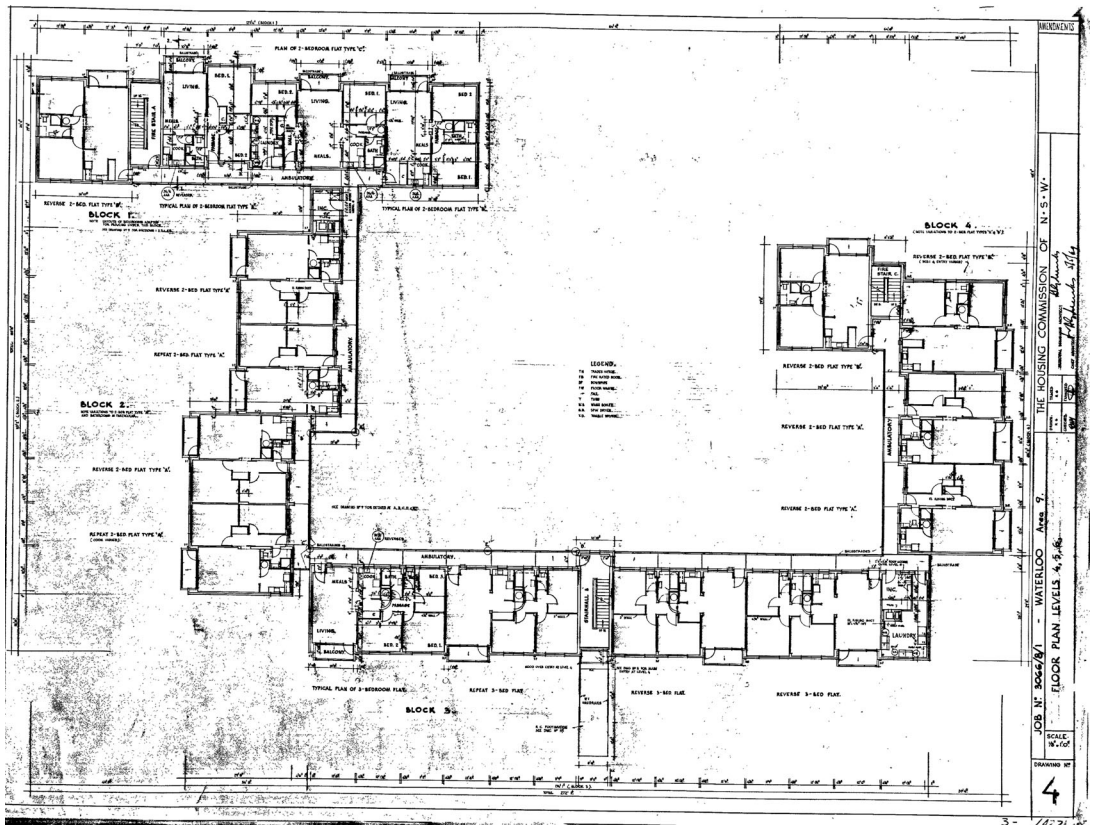



**Figure 9.** Camelia Grove floor plans showing long 'ambulatory' access around courtyard.

providing ample natural daylight and ventilation, and well-constructed of robust materials that have stood the test of time.

While most of the estate is quite flat, the southeastern corner rises steeply and creates challenges for building in those sections. This change in levels would have made the laying out of multiple free-standing buildings with small footprints – of the kind produced on the estate up until that time – a complex exercise. The steep gradient created difficulties for construction and resident access. To address these challenges, the Commission looked beyond their regular in-house design team and sought input from architects in private practice. Lipson, Kaad and Fotheringham designed Madden Place in 1966. It was composed of one larger, single building with 56 dwellings that could respond to the demanding particularities of the site. In 1969 the Housing Commission Special Designs Group were the architects for Camelia Grove, with 65 dwellings, and, like Madden Place, it was configured to respond directly to the underlying contours. Both are large C-shaped courtyard buildings in plan that are serviced by long open corridors around their internal perimeter. They are varied in height, reducing from six to four storeys as the land beneath them rises (Figures 9 and 10). The real innovation in their design is in section where the buildings are arranged as two three-storey walk-ups stacked on top of each other, with on-grade access from the lowest level and access to the fourth level via a bridge from the higher street frontage. This arrangement delivered compliant fire egress while avoiding the need for elevators, which the

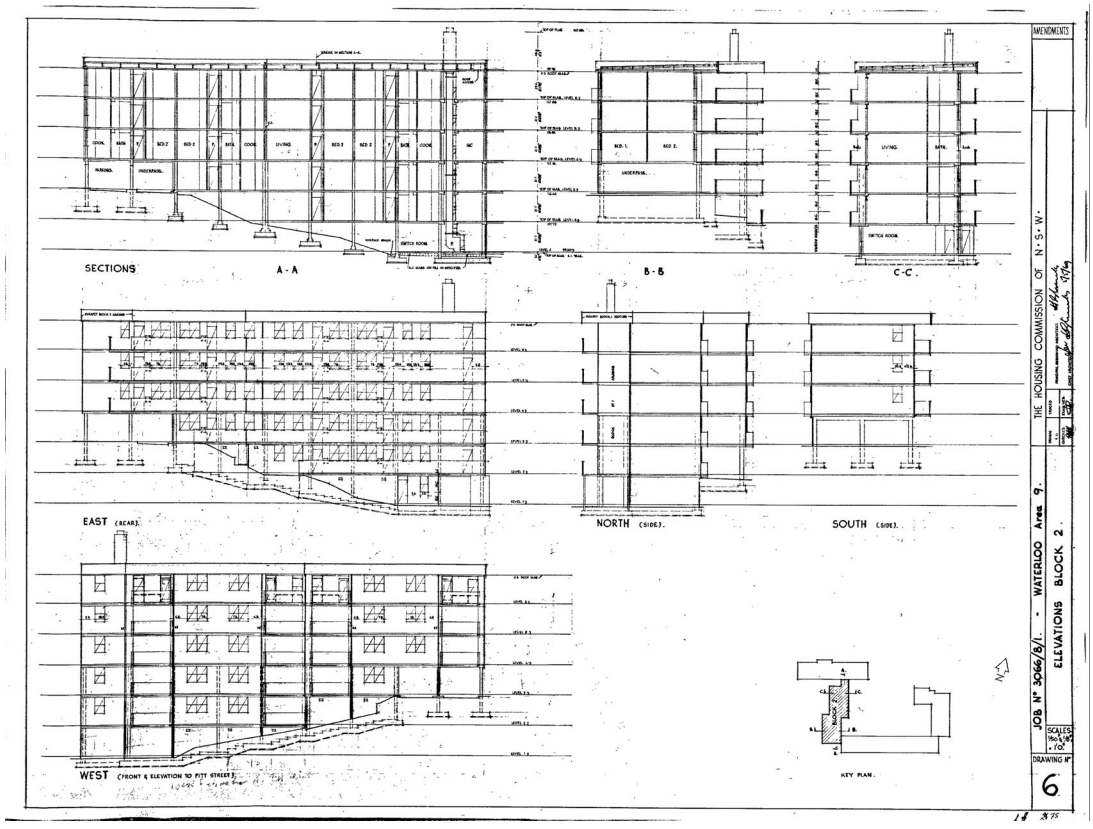



**Figure 10.** Camelia Grove elevations and sections showing relationship to steep topography.

Commission generally tried to avoid in its buildings, being too expensive at the time as well as being difficult to maintain.

Camelia Grove and Madden Place are much larger than buildings that had been constructed earlier on the estate, making for larger groupings of households with up to sixteen units per floor. A consequence of this arrangement was that many units have only one uncompromised orientation for outlook and visual privacy, with their second orientation facing back to the corridor, often for the window to the second bedroom. For some units, this configuration also means that the living room and balcony faces south and does not receive direct sunlight. Due to the increased number of dwellings, shared laundries and 'incinerator' or waste rooms are provided on each floor rather than as a single room at ground floor in the garden areas like the smaller buildings. Internally, units adopt a conventional and compact layout not unlike the earlier buildings and meeting similar standards in terms of space provision and facilities provided. These blocks still have comparably narrow building depths, especially compared with most of today's residential development in Australia, and include large windows for light and air.

In the 1970s, the ambitions of the Housing Commission for the estate dramatically increased. Larger projects in nearby Redfern such as William McKell Place by Morrow & Gordon Architects in 1961 (8–10 storeys, 186 dwellings) and Poets Corner by Peddle, Thorp & Walker Architects in 1966 (three 17-storey slabs with 192 dwellings each) set the precedent for the final stage of

Waterloo.[22] The Endeavour Project in the northern portion of Waterloo, built between 1970 and 1976, comprised four 17-storey slabs (named James Cook, Joseph Banks, Daniel Solander and Marton) each with 214 units, and two 30-storey towers (Matavai and Turanga) each with 232 units and designed specifically for aged pensioners (Figure 11). This project became the most visible culmination of the Housing Commission's activities in New South Wales and possibly the most prominent in the public imagination. Just under 300 dwellings, amongst other buildings, were demolished and several streets and lanes subsumed to make way for the Endeavour Project. All six buildings were set in expansive open space, surrounded by car parking and park-like lawned spaces, with building entries relatively distant from the street.

The repetitive nature of tall building floor plates coupled with pre-cast concrete construction technology allowed for rapid building at scale and was able to utilize the provision of lifts to their fullest. The typical floor of the cruciform-plan towers includes a north-facing community room at each lift lobby, south-facing laundry, drying and garbage rooms, and eight small units arrayed around a dual lift core and 'scissor' fire stairs. Seven of the eight units on each floor are bedsit flats comprised of a kitchen, bathroom and single combined space for living and bedroom with windows to a single orientation north, east or west. The eighth is a one-bedroom unit. South-facing units were given windows in a second direction providing some sunlight access, but no units had balconies and ceilings are relatively low. At the ground floor, a lift lobby, letterbox area, office, sitting room, and community room contribute additional shared facilities. Externally, the provision of a generous fenced landscaped area with seating, shelter, and water features defined a secure communal garden (Figure 12).

The four slab blocks run north to south, with units facing either east or west, which maximizes solar exposure and makes westward-facing units especially subject to intense afternoon solar gain and overheating. The strictness of construction efficiencies does not appear to have allowed for windows to face north or south at end units and none of the units enjoy proper, natural cross-ventilation. The units are served by long, narrow, highly internalized corridors with turns at the lift lobby preventing a straight line of view from the lift to the unit entry. A laundry and garbage chute are located at each end of the floor plate for convenience. The double-loaded corridor arrangement means that the building section is relatively deep compared to previous buildings but would still be considered very slender by today's standards which can typically be almost double, up to 24 m. Almost all kitchens and bathrooms were located internally along the back wall of each unit, requiring artificial lighting and mechanical ventilation. Windows to bedrooms were wide, but with relatively high sills and low head heights. This is seemingly intended to create a feeling of safety at height but limits a good outlook. Taller sliding glass living room doors to small squarish balconies provide some additional glazing but visual access to outside is diminished by a solid balustrade detail (Figures 13 and 14).

In 1972, the Housing Commission announced plans to extend the site to the east and build more than two thousand more dwellings in Waterloo, including up to six more 30-storey towers to a different design. This was opposed by the newly formed South Sydney Action Group.[23] In 1973, the Commonwealth intervened in State and Local housing provisions through its Department of Urban and Regional Development with a change in policy to protect and rehabilitate inner-city suburbs of terrace housing in places such as Glebe and Woolloomooloo.[24] In 1974, the Builders

[22]Zanardo and Boyd, "Affordable Housing Sydney".
[23]Burgmann and Burgmann, *Green Bans, Red Union*, 148.
[24]Milligan, "How Different?" 94.




**Figure 11.** Matavai and Turangai as seen on the cover of a Housing Commission of NSW publication about Waterloo.

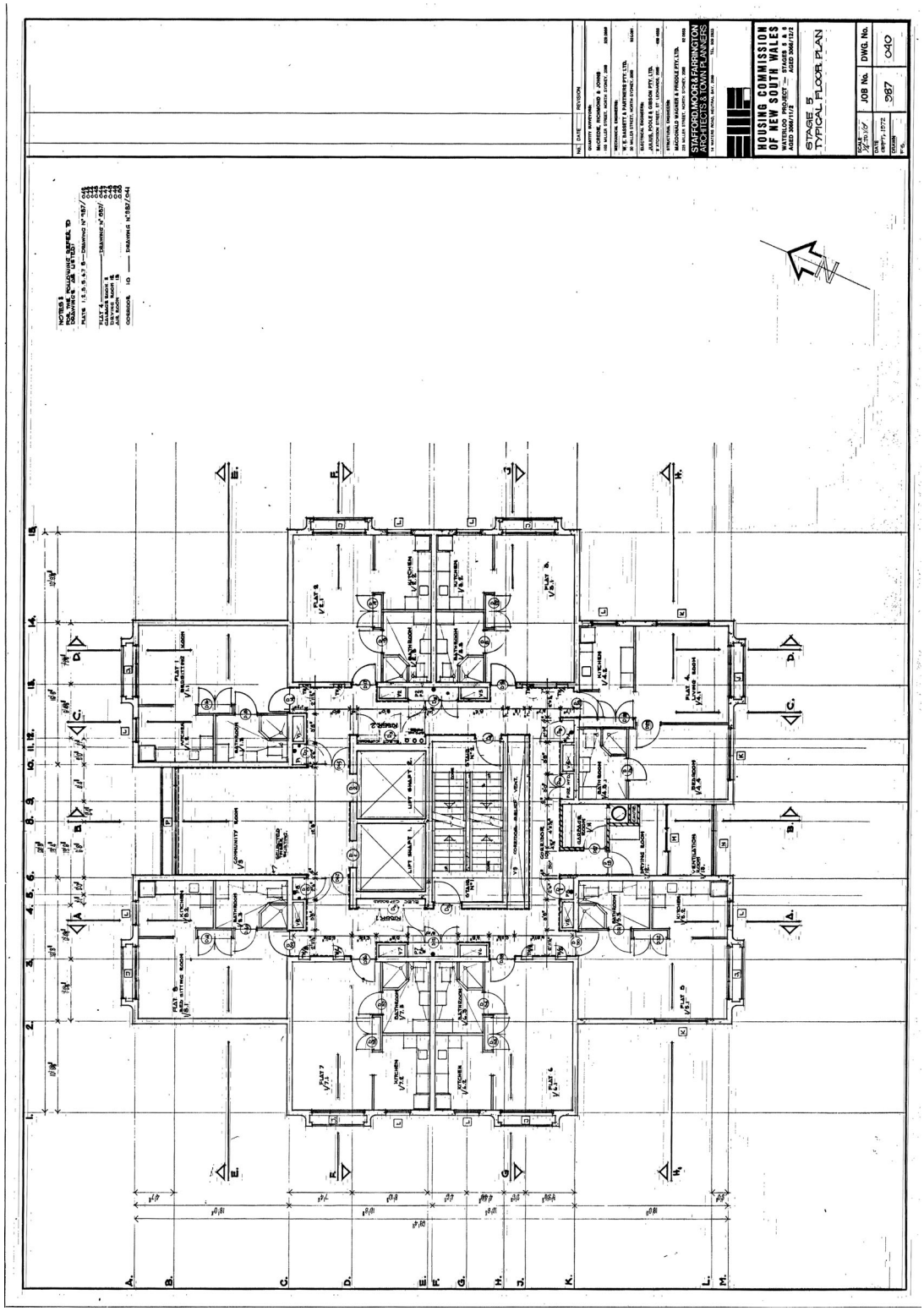



**Figure 12.** Matavai plan with community room at lift facing north, shared laundry and waster room services centrally to south and bedsit unit layout.

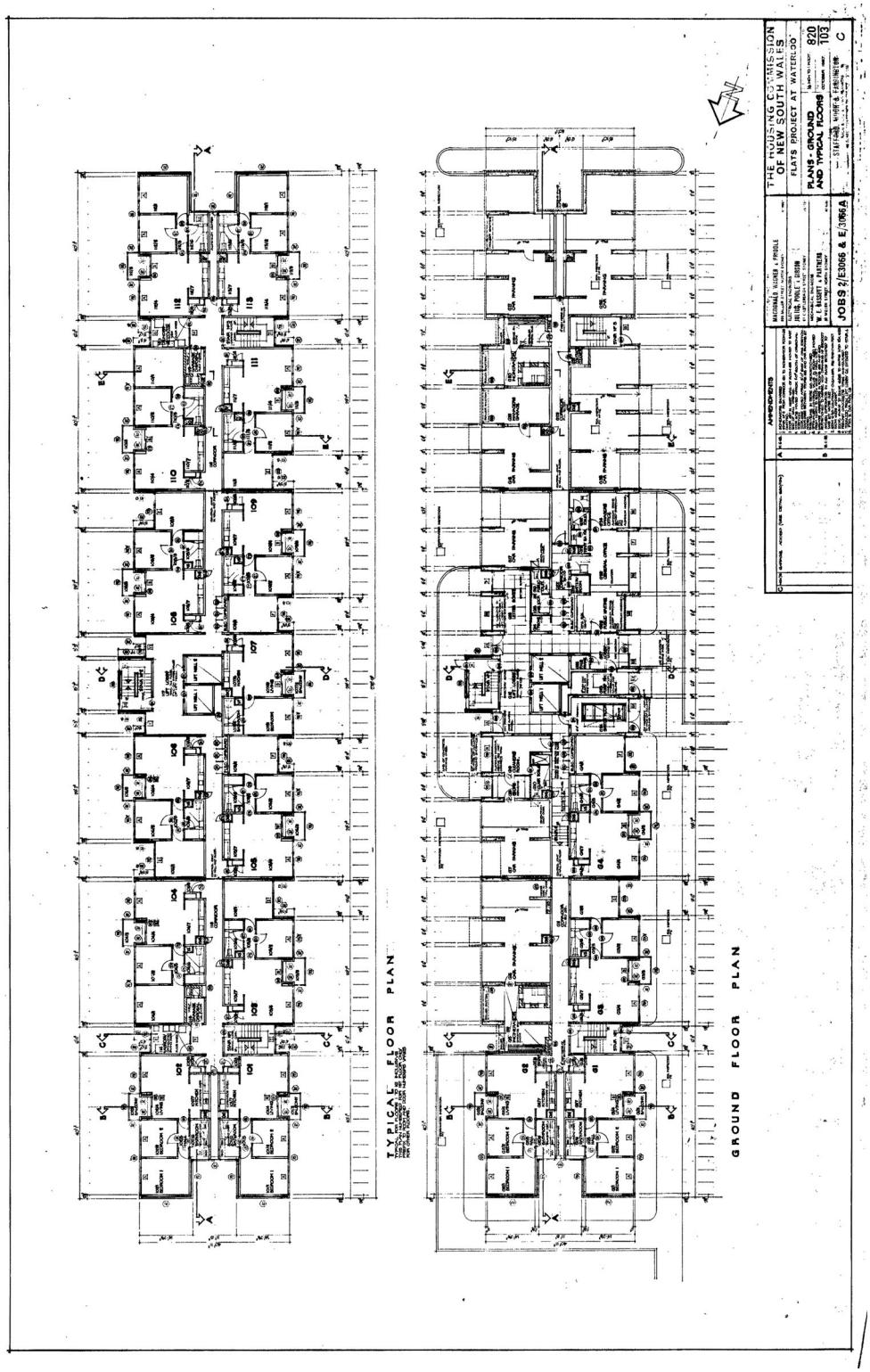



**Figure 13.** Plan of James Cook ground floor and typical floor showing long internalized corridors with double-loaded unit arrangement.



**Figure 14.** Photo of James Cook showing effectively blank northern end wall and eastern facades with small fenestration and solid balconies.

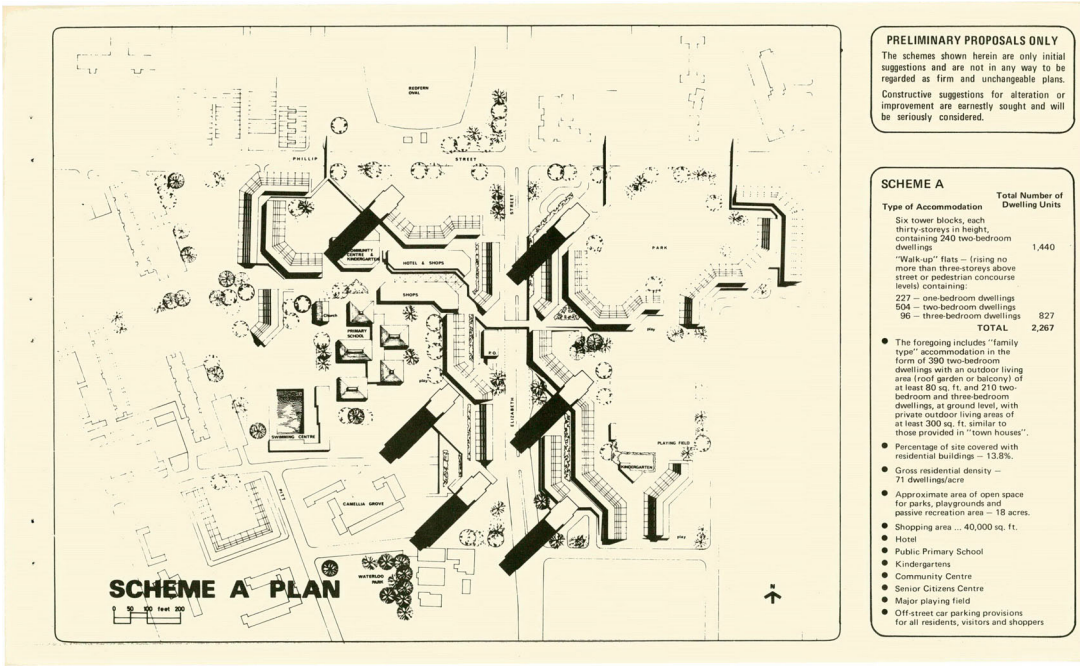



**Figure 15.** Unrealized plan for eastern part of Waterloo with six more towers to the height of Matavai and Turanga.

Labourers Federation placed a 'green ban' on further demolition of low-income housing in Waterloo. In response, the Housing Commission commenced a detailed consultative process with residents resulting in their activity being redirected towards an extensive programme of rehabilitation of existing housing complimented by architecturally compatible new infill housing (Figure 15).[25]

More dwellings were added to the southern end of the estate right up until the early 1980s, including the architecturally striking Drysdale and Dobell developments (also known as 'Artists Corner') designed by Tao Gofers and Penny Rosier of the NSW Housing Commission. But in many ways, the green ban and federal engagement in the mid-1970s decisively shifted the trajectory of public housing development in Sydney. The quarter century of expansive growth that created the Waterloo estate was over.

It had been a period of energetic ambition and growing capacity for public housing that left a diverse built legacy, most of it still physically intact and still inhabited. But how should we understand and evaluate that legacy? If we were to listen to residents who live in the buildings we have just described, it is quite likely that the current plans for estate redevelopment would not go ahead. Similarly, if we look to the NSW state government's own measures for what constitutes good housing, the options for improving the estate's housing stock appear quite different to what have been presented thus far. The government has not publicly presented a systematic analysis of how the housing on the Waterloo estate measures up against their own standards. Using a series of Wellbeing Criteria that are discussed in the next section, we have attempted to do this and presented the results in Table 1.

[25]Butler-Bowdon and Pickett, *Homes in the Sky*, 141.

**Table 1.** Analysis of Waterloo Housing Type Amenity.

| WELLBEING CRITERIA | ARCHITECTURAL QUALITY | CONTROL REQUIREMENT | SUPPORTING POLICY | 3-STOREY WALK-UPS (BF822D Block 2) | 6-STOREY COURTYARDS (Camellia Grove) | 17-STOREY SLABS (James Cook) | 30-STOREY TOWERS (Matavai) |
|---|---|---|---|---|---|---|---|
| Natural light | Solar access | Minimum of 70% of units receive minimum of 2 hours of direct sunlight to living rooms and private open spaces between 9am and 3pm at mid-winter. | ADG 4A-11 | ✗ (too many south-facing) | ✓ | ✓ | ✓ |
| | | Maximum of 15% of units receive no direct sunlight to living rooms and private open space between 9am and 3pm at mid-winter. | ADG 4A-13 | ✗ (too many south-facing) | ✗ (too many south-facing) | ✓ | ✓ |
| | | Neighbouring properties should receive equivalent minimum solar access. | ADG 3B-2 1 | ✓ | ✓ | ✓ | ✓ |
| | Windows size and location | Minimum window glass area must be 10% of floor area of the room. | ADG 4D-1 2 | ✓ | ✓ | ✓ | ✓ |
| | | Daylight may not be borrowed from other rooms. | | ✓ | ✓ | ✓ | ✓ |
| | | A window should be visible from any point in a habitable room. | ADG 4D-1 4 | ✓ | ✓ | ✓ | ✓ |
| | Habitable room depth | Maximum 8m from a window for open-plan layouts. | ADG 4D-2 2 | ✓ | ✓ | ✓ | ✓ |
| | Ceiling height | Minimum 2.7m floor-to-ceiling in habitable rooms. | ADG 4C-1 1 | ✗ (~8′4″/ 2.54m) | ✗ (~8′4″/ 2.54m) | ✗ (~8′4″/ 2.54m) | ✗ (~8′4″/ 2.54m) |
| Fresh air | Natural cross ventilation | Minimum of 60% of units are naturally cross ventilated in the first nine storeys. | ADG 4B-3 1 | ✓ | … (through open gallery) | ✗ (double-loaded corridor) | ✗ (single-aspect units) |
| | Windows size and location | Minimum unobstructed window openable area must be 5% of floor area of the room. | ADG 4B-1 3 | ✓ | ✓ | ✓ | ✓ |
| | | Air may not be borrowed from other rooms. | ADG 4D-1 2 | ✓ | ✓ | ✓ | ✓ |
| | Habitable room depth | Maximum 8m from a window for open-plan layouts. | ADG 4D-2 2 | ✓ | ✓ | ✓ | ✓ |
| | Ceiling height | Minimum 2.7m floor-to-ceiling in habitable rooms. | ADG 4C-1 1 | ✗ (~8′4″/ 2.54m) | ✗ (~8′4″/ 2.54m) | ✗ (~8′4″/ 2.54m) | ✗ (~8′4″/ 2.54m) |
| Visual PRIVACY acoustic Privacy | Building separation | Up to 4th storey, habitable room windows and balconies minimum 6m separation from a side or rear boundary (3m non-habitable room windows). | 3F-1 1 | ✓ | ✓ | ✓ | ✓ |
| | | For the 5th–8th storey, habitable room windows and balconies minimum 9m separation from a side or rear boundary (4.5m non-habitable room windows). | | | | | |

(*Continued*)

**Table 1.** Continued.

| WELLBEING CRITERIA | ARCHITECTURAL QUALITY | CONTROL REQUIREMENT | SUPPORTING POLICY | 3-STOREY WALK-UPS (BF822D Block 2) | 6-STOREY COURTYARDS (Camellia Grove) | 17-STOREY SLABS (James Cook) | 30-STOREY TOWERS (Matavai) |
|---|---|---|---|---|---|---|---|
| | | For the 9th storey and above, habitable room windows and balconies minimum 12m separation from a side or rear boundary (6m non-habitable room windows). When separation is between two buildings on the same site combine required separations. | | | | | |
| | Ground floor apartments | Provide changes in level, landscaping and front fences to achieve visual privacy. | ADG 3C-1 2 | …(varied) | ✓ | ✓ | ✓ |
| | Relationship to circulation | Primary living and bedroom windows should not open onto common circulation spaces. | ADG 4F-1 10 | ✓ | ✗ (minor bed to gallery) | ✓ | ✓ |
| | Balustrades | Provide solid or partially solid balustrades to achieve visual privacy from the street. | ADG 4E-3 1 | ✗ (open balustrades) | ✓ | ✓ | n/a |
| Community | Communal open space | Minimum of 25% of site area as communal open space. | ADG 3D-1 1 | ✓ | ✓ | ✓ | ✓ |
| | | Minimum 2 hours direct sunlight to 50% space between 9am and 3pm at mid-winter. Minimum 3m wide. | ADG 3D-1 2<br>ADG 3D-1 4 | ✓<br>✓ | ✓<br>✓ | ✓<br>✓ | ✓<br>✓ |
| | Address | Building entries should be clearly identifiable and relate to the street network. | ADG 3G-1 2<br>ADG 3G-1 3 | (visible but distant) | (visible but distant) | ✗ (distant through parking) | … (from driveway) |
| | Number of units | Maximum 8 units off a circulation core on a single level. | ADG 4F-1 1 | ✓ | ✗ (up to 16 on single level) | ✗ (up to 13 on single level) | ✓ |
| | Circulation areas | Daylight and natural ventilation to all circulation spaces. | ADG 4F-1 4 | ✓ | ✓ | (largely internalized) | ✓ |
| Nature | Deep soil zones | Minimum of 7% of site as deep soil. Site area less than 650 $m^2$ any width/650 $m^2$–1500 $m^2$ at 3m wide/1500 $m^2$ at 6 m wide. | ADG 3E-1 1 | ✓ | ✓ | ✓ | ✓ |
| | Tree planting | Site area up to 850 $m^2$ 1 medium tree per 50 $m^2$ of deep soil/Over 850 $m^2$ 1 large tree or two medium trees per 90 $m^2$ of deep soil. | ADG 4O-1 4 | ✓ | ✓ | ✓ | ✓ |
| Accessibility Livability | Unit size | Minimum Studio 35 $m^2$/1 bedroom 50 $m^2$/2 bedroom 70 $m^2$/3 bedroom 90 $m^2$ Additional bathrooms +5 $m^2$/Fourth and additional bedrooms +12 $m^2$. | ADG 4D-1 1 | ✗ (1B ∼45m2 / 2B ∼55m2) | ✗ (2B ∼57m2 / 3B ∼77m2) | ✗ (2B ∼55m2) | ✗ (St ∼25m2 / 1B ∼33m2) |
| | Bedroom size | Minimum 3 m dimension plus wardrobe. | ADG 4D-3 2 | ✗ (minor bed 9′3″ / 2.82m) | ✗ (minor bed 8′5″ / 2.57m) | ✗ (minor bed no robe) | n/a |
| | Living room width | Minimum Studio and 1 bedroom living width 3.6 m/2 bedroom and 3 bedroom living width 4 m. | ADG 4D-3 3 | ✓ | ✗ (2 & 3 bed 12′6″ / 3.81m) | ✓ | ✗ (St ∼9′3″ / 2.82m) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | Balcony size | Minimum Studio 4 $m^2$ at 1 m wide/1 bedroom 8 $m^2$ at 2 m wide/2 bedroom 10 $m^2$ at 2.4 m wide/3 bedroom 12 $m^2$ at 2.4 m wide/Ground level or podium 15 $m^2$ at 3 m wide. | ADG 4E-1 1<br>ADG 4E-1 2 | ✗ (many with no balconies) (∼3m2 at2′9″ / 0.84m wide) | ✗ (∼4m2 at 3′9″ / 1.14m wide) | ✗ (∼4m2 at 6′ / 1.82m wide) | ✗ (no balconies) |
| | Storage | Minimum Studio 4 $m^3$/1 bedroom 6 $m^3$/2 bedroom 8 $m^3$/3 bedroom 10 $m^3$ with at least 50% volume provided within unit. | ADG 4G-1 1 | ✗ (minimal built-in storage) | ✗ (minimal built-in storage) | ✗ (minimal built-in storage) | ✗ (minimal built-in storage) |
| | Universal design | Minimum of 20% of units designed to Silver Level Livable Housing Design Guidelines. | ADG 4Q-1 1 | ✗ (spatially inadequate) | ✗ (spatially inadequate) | ✗ (spatially inadequate) | ✗ (spatially inadequate) |
| Thermal comfort | Shading | Windows incorporate sun shading. | ADG 4A-3 1 | ✗ (no external shading) | ✗ (no external shading) | ✗ (no external shading) | …(minimal external shading) |

## Evaluating Waterloo

There has been a growing effort in recent years to systematize and build an evidence base for good architectural design. This has partly been driven by a desire to influence design guidance used by governments and third sector actors to promote standards of environmental comfort and other factors perceived as supportive of users' physical and psychological wellbeing. Much of the initial research for evidence-based design was developed for health settings, moving beyond the high modern concept of the hospital as a curative apparatus and provide settings to support the overall wellbeing of patients, and has since extended into a range of other architectural settings and situations. Researchers have demonstrated that views of and access to green space, high standards of air quality, and thermal comfort levels all have a direct impact on health. Well-designed buildings minimize mould and dampness, maintain comfortable air quality, and provide spaces for outdoor recreation and exercise. These qualities have also been shown to contribute to a reduction in the spread of infectious diseases, as the COVID-19 pandemic has recently highlighted, while supporting better cardiovascular and respiratory health.[26]

Many of these indoor environmental and spatial qualities have long been recognized by architects as fundamental biological and psychological factors that support human flourishing. A growing literature focusing on the idea of wellbeing has confirmed that many of these qualities – particularly thermal comfort and access to sunshine and green spaces – have an impact on psychological wellbeing. Other design considerations that become important in the context of psychological comfort include the provision of privacy and opportunities for social connection. General principles arising from decades of academic and industry research provide a basis for policymakers to arrive at a sense of what 'best practice' should look like, including for housing.[27]

In NSW, a combination of physical and psychological wellbeing principles are effectively captured in the *Apartment Design Guide* (2015) planning policy which establishes a benchmark of the quality that must be achieved in the construction of any new residential apartment development that is at least three storeys in height and contains at least four dwellings. While this document was created with the private housing market in mind, it is nonetheless applicable to all tenures. In the context of the Waterloo estate, it provides a useful instrument for testing claims that its buildings have reached the end of their usable lifespan or are no longer fit for purpose, as it provides a contemporary measure of what is considered acceptable in terms of overall amenity in new apartments.

To establish this benchmark, we thematically analysed the *Apartment Design Guideline* relative to the environmental factors known to have the greatest impact on resident amenity, and, therefore, their overall wellbeing (Table 1). Relative to the provision of natural light, fresh air and thermal comfort, the guideline sets expectations about how much direct sunlight, daylight, natural ventilation and natural cross ventilation should serve an apartment, stipulating window orientation, openability, size (as a proportion of a room's floor area) and shading, as well as setting minimum ceiling heights and maximum habitable room depths. Recommendations to support visual and acoustic privacy set expectations regarding the separation between buildings, and the provision and placement of balconies, windows, and landscaping. Under the theme of liveability, we have grouped considerations related to physical accessibility and minimum areas and dimensions for unit, room, and balcony sizes. While considerations of room size are difficult to measure and have thus not been the focus of design-related research relative to wellbeing in residential

[26]Hood, "Dwelling Disparities," A310–A317; Kent et al., "Healthy Built Environments".
[27]Baum and Davis, "Reducing the Stress of High Density Living"; Engel et al., "Older Adults' Quality of Life"; Evans et al., "Housing Quality and Mental Health," 526; David Halpern, "Mental Health and the Built Environment".

environments, at least one study of tenants relocated within the renewal process of an Australian housing estate found that the provision of larger rooms was a significant contributor to tenant satisfaction within those relocations; this suggests a link between more generous interior spaces and psychological comfort.[28]

Design considerations aimed at supporting a sense of community include expectations around the size and quality of communal spaces; the design of building entrances to be easily identifiable and maintain a clear relationship to the street; and the maximum number of apartments per floor accessible from a circulation core. While design is not the main determinant in establishing a strong sense of community, spaces we might term *uncomfortable* can have a negative psychological impact, particularly around perceptions of discomfort and danger. Such spaces, in turn, can impede communication and a person's resilience to irritation – both influencing a tenant's likelihood of establishing and maintaining social contact with neighbours.[29]

In the remainder of this section, we assess the four housing types on the estate against four key types of amenity that broadly capture the objectives of the *Apartment Design Guide*. An analysis of the four main housing types contained within the Waterloo Estate – 3-storey walk-ups, 6-storey courtyards, 17-storey slabs, and 30-storey towers – confirms that some typologies go further in meeting contemporary standards than others (Table 1). It must also be acknowledged that design is a practice of multiple possibilities where balancing the many and sometimes conflicting requirements of any building can result in necessary compromises in some areas to optimize benefits in others. This is true of new construction that aspires to meet contemporary guidelines as much as it is of established apartments. It should also be acknowledged that not all design considerations discussed have an equal impact on wellbeing.

All four accommodation types compare well against the *Apartment Design Guide* recommendations in relation to access to light and air, and thermal comfort. The minor compromise across all four types is that ceiling heights are only 2.5 m, which met standards at the time of construction but is 200 mm less than the current design criteria. It is also worth noting that this is still greater than the 2.4 m ceiling height that remains standard in other types of residential construction. Due to building placement and orientation on site, the 3-storey walk-ups have less access to sunshine than the other housing types, with less than half of dwellings able to receive a minimum of 2 h of direct sunlight between 9am and 3pm at mid-winter. However, this compromise may be partially mitigated by the ease of access to substantial outdoor greenspace from this housing type. In respect of the 17-storey slabs and 30-storey towers, the inclusion of double-loaded corridors and/or single-aspect units present a much larger compromise relative to enabling the cross-ventilation of these units (noting that natural cross ventilation is deemed-to-comply above the ninth storey due to greater wind speeds at height). None of the four building types appear to provide adequate sun shading to windows, although this could be easily addressed through retrofitting externally.

In respect to visual and acoustic privacy, all four accommodation types compare well against the *Apartment Design Guide* with only minor issues arising. These include the open balustrades (instead of solid for privacy) in the case of the 3-storey walk-ups; and the occurrence of a second bedroom occasionally facing an external circulation corridor in the case of the 6-story courtyard building type.

[28]Baker and Arthurson, "Housing, Place or Social Networks," 28–35.
[29]Halpern, *Mental Health and the Built Environment*, 133–54. Kleeman et al., "The Impact of the Design and Quality of Communal Areas," 10.

Access to nature and a sense of community seem like harder amenities to measure. But the design guide contains provisions that are broadly supported by research. The recommendation that a maximum of 8 residences should be accessed from a circulation core is an important consideration, as research shows that encountering a smaller and known group of neighbours enhances feelings of psychological comfort and physical safety.[30] In the 6-storey courtyards and 17-storey slabs, circulation cores are shared by between 13 and 16 residences, in the case of the latter, potentially doubling what is recommended by *Apartment Design Guide* as a comfortable level of social encounter. This discomfort is further exacerbated in the 17-storey slabs owing to the lack of natural light and ventilation available to those circulation spaces which are long, tight and largely internalized. While there are limited options for improving this situation in the 17-storey slabs, the retrofitting of lift cores to the 6-storey courtyard buildings has been proven elsewhere to be a feasible solution. In the case of the 6-storey courtyard buildings this could allow them to effectively become multiple attached buildings, permitting some units to cross-through the building which could improve natural cross ventilation as well as improve privacy to second bedrooms. The inclusion of lift cores would also assist these walk-up buildings to become accessible. Building entrances are another design consideration related to community and, for all four accommodation types, represent an aspect of design that would welcome some improvement with some design attention given at the street edge.

Across the four housing types, unit sizes are between 10% and 30% smaller that the *Apartment Design Guide* recommends. This equates to a shortfall of 13–15 $m^2$ across the 2- and 3-bedroom apartments for all building types, and a shortfall of 17 $m^2$ to the 1-bedroom units in the 30-storey tower (noting that a significant minority of 1-bedroom units were consolidated to form 2-bedroom apartments around the turn of the century). Across all four housing types, balconies, where provided, are often inadequate in terms of overall dimensions relative to the design criteria; while none of the four accommodation types provide accessibility features to meet Silver Level of the Liveable Housing Design Guidelines such as a step-free environment, door and corridor widths and suitable bathroom configuration. Built-in storage within apartments is also lacking as a consequence of smaller unit sizes.

While this analysis highlights some failings, what it in fact identifies is that much of the existing housing at the Waterloo Estate has fundamental design attributes that are supported by evidence and promoted by the NSW Government in its own endorsed contemporary design guidance. All dwellings have large windows to provide excellent daylight and fresh air to all dwellings; rooms are never too deep; and there is generous open space between and around the buildings with abundant deep soil area allowing for the planting of large trees. Despite requiring various additions and upgrades – some more critical than others – the idea that they are definitively inappropriate for contemproarry needs and, therefore, obsolete is simply not supported by the evidence.

Design considerations are never black and white, particularly given that architects and decision-makers have access to design-based evidence that remains limited at best. For example, there is no definitive way of assessing the benefits of expansive city views available from apartments in the 30-storey towers. Do those benefits outweigh the drawbacks of small apartment plans in the tallest towers on the Waterloo Estate? These apartments were all originally smaller than current NSW policy prescribes, and while some apartments have been consolidated to create larger two-bedroom units, most remain smaller than the guidelines recommend. Equally, it is reasonable to ask whether

[30]Baum and Davis, "Reducing the Stress of High Density Living,"478–80; Evans et al., "Housing Quality and Mental Health," 526; Halpern, *Mental Health and the Built Environment*, 152–4.

shared living spaces on each floor of the 30-storey buildings compensates for that deficit. Furthermore, we do not have methods to accurately quantify the impact on wellbeing of living with a ceiling that is 200 mm lower, or a balcony that falls below the minimum of 4 m × 1 m wide (in the case of a studio apartment) – two aspects where these apartments fall short of the Apartment Design Guide. Importantly, these small compromises should be considered in relation to the social disruption and upheaval of forced relocation.[31] The possible adverse health outcomes are a particular concern for older residents of these buildings who are in general the most vulnerable in this respect and least likely to benefit from redevelopment.[32]

It is clear then that while the buildings on the Waterloo Estate have some deficiencies, the idea that they are definitively not fit for purpose is simply not true. With the possible exception of the slab blocks, most of the buildings on the estate could have their lives usefully extended and continue to serve their purpose in a way that supports residents with relatively modest renovation, or even with additions. With effort, even the slab blocks could be substantially improved through staged upgrades that would still be much less disruptive than what the NSW Government currently has in mind. Some upgrades of this kind have already happened. Units in the Matavai and Turanga towers, for example, have been combined. We also know that in addition to being less disruptive for residents, such retrofit and transformation strategies help to avoid the use of new materials and the emissions associated with them as well as reducing the amount of waste that goes to landfill as part of clearance-led renewal schemes.[33] Unquestionably housing agencies and architects should be broadly pursuing the different forms of amenity that support resident wellbeing to the greatest extent possible. But equally, the question remains, does the government's plan properly prioritize resident wellbeing? One would expect that the government has thoroughly assessed the risks to inhabitants as well as the projected benefits if it were primarily motivated by the interests of public housing tenants. An analysis of the discourse around the redevelopment case put publicly by government, however, points to other motivations.

## Rationalizing redevelopment

While improving housing quality is an objective for the Waterloo redevelopment, it was just one of several that motivated the project and arguably not the most powerful. Consistent with the redevelopment of public housing estates throughout many parts of the Global North, the redevelopment of Waterloo was also driven by the desire to deconcentrate socio-economic disadvantage and transform a stigmatized district into one seen to be more befitting its location in the highly coveted inner city of Sydney.[34] Equally, it was a response to funding arrangements which severely constrain the capacity of housing authorities to expand their housing stock and compel them to redevelop existing landholdings via partnerships with the private sector, with the sale of new market-rate housing cross-subsidizing the delivery of social housing.[35]

The Waterloo project was devised as part of the Communities Plus program of estate redevelopment, which included 'major projects' targeting large estates with high land values, close proximity

[31]Porter et al., *Understanding the Drivers*.
[32]Baker and Arthurson, "Housing, Place or Social Networks," 32–3; Crawford and Sainsbury, "Opportunity or Loss?"; Engel et al., "Older Adults' Quality of Life".
[33]Architecture 2030, *The CARE (Carbon Avoided: Retrofit Estimator) Tool*.
[34]Darcy, "De-concentration of Disadvantage"; August, "Revitalisation Gone Wrong"; Goetz, "The Politics of Poverty Deconcentration"; Watt, "Territorial Stigmatisation and Poor Housing".
[35]Groenhart and Burke, *Thirty Years of Public Housing Supply*; see also, Darcy and Rogers, "Place, Political Culture and Post-Green Ban Resistance"; Lees, "The Urban Injustices".

to employment and economic growth areas, good transport connections and, importantly, a history of relatively low-density zoning allowing for significantly intensified development.[36] As recent work by Wynne et al. outline, the programme signalled the increasing relevance of land values and development yield to the selection of renewal sites, as well as to wider asset disposal strategies.[37] A key component of the Future Directions for Social Housing strategy of the Liberal-National coalition government of the time, was that the programme responded to the long-term decline in Commonwealth funding and absence of state government funding by replacing and slightly increasing social housing stock at no net cost to the NSW Government. In essence, private property development on redeveloped estates would fund the replacement of, and a slight increase in, social housing. Rather than public housing managed by the state housing authority, this social housing would be managed by community housing organizations, capitalizing on those organizations' exemption from paying Goods and Services Tax and their tenants' eligibility for Commonwealth Rent Assistance (and thereby shifting the cost of housing subsidies from state to Commonwealth government).[38]

Waterloo comfortably met the Communities Plus criteria, as an 18-ha estate just a few kilometres from the Sydney CBD as well as University of Sydney, University of Technology Sydney, and Australian Technology Park. It is also only a kilometre or so from Redfern Station – a major interchange on Sydney's railway network. Despite this, the NSW Government decided to locate a new Metro station at Waterloo, which had the effect of further enhancing land values and ensuring financial feasibility according to the Communities Plus constraints. The redevelopment of the Waterloo estate was announced with a letter to tenants from then-Minister for Social Housing, Brad Hazzard, stating:

> I am excited to write to you and let you know Transport for NSW has announced a new Sydney Metro station at Waterloo, due to be completed by 2024.
>
> On my visits to Waterloo I have spoken to a number of tenants who have told me they like living in the area, but are disappointed with the quality of the ageing housing on the Waterloo estate.
>
> The new Metro station not only brings fantastic new rapid transport to Waterloo, it means we can redevelop the estate with brand new and more social housing that meets people's needs.
>
> […]
>
> The Metro station and the area's renewal will transform the Waterloo housing estate for the better, building a dynamic new community with great transport, better homes, better facilities and more jobs.[39]

While Hazzard's letter implied that the decision to locate a Metro station in Waterloo catalysed the redevelopment of the estate, the *Sydney Morning Herald* reported that it was the government's desire to redevelop the estate that led to the decision about the Metro station's location. According to that story, Ministers for Social Housing, Planning, and Police, as well as representatives of the state-owned Urban Growth Development Corporation, were advocating for a Waterloo Metro station as a catalyst for the project. These four voices persuaded then-Premier Mike Baird to cast his deciding vote on the NSW Cabinet Infrastructure Committee for Waterloo, ahead of an

[36]Land and Housing Corporation, "Communities Plus Industry Briefing".
[37]Wynne et al., "The Marketisation of Social Housing"; Ruming and Melo Zurita, "Care and Dispossession".
[38]Sisson, "Public Housing and Territorial Stigma".
[39]Hazzard, "Letter from Minister Hazzard to Waterloo Tenants".

alternative station location at the University of Sydney. In addition to the opportunity to redevelop state-owned housing at little or no cost to government, the project was viewed as a vehicle for increasing inner-city housing supply, and a means of addressing perceived social problems in the environs of the estate related to concentrated poverty.[40]

There is growing pressure and desire in NSW to increase housing density in inner city locations and private housing supply in general.[41] While there are sound social and environmental arguments for densifying inner urban areas, it is also notable that significant government revenue is generated through property transfer duties and land taxes; that the then-government undertook a significant 'asset recycling' programme involving revenue generation and reinvestment through privatization of public infrastructure, land, and services; and that the property development and real estate industries have exercised a powerful influence on policy and planning decisions in NSW historically.[42] Furthermore, in a planning system where government mostly is unable to do more than regulate the land use options of private property owners, large plots of public land like the Waterloo estate are rare 'low hanging fruit' for realizing these objectives, irrespective of the negative social impacts of redevelopment. The redevelopment of the Waterloo estate would add as many as 6000 new private apartments in an area seen as insufficiently dense for its proximity to the Sydney CBD.

Recent political leaders in NSW and their policy advisers have subscribed to the idea that large areas of concentrated public housing produce negative social outcomes due to their concentration of social disadvantage and so-called 'neighbourhood effects' – a concept which contends that a high incidence of disadvantage in a local area is a cause of further disadvantage for the individuals and communities who live there. A detailed discussion of this highly contested concept is outside the scope of this paper. Suffice to say, it is an idea that, as Joseph Heathcott has noted, has been deployed by architects, sociologists, and planners alike and seized upon by political decision-makers to justify the retrenchment of public housing.[43] For instance, in making the case for the Waterloo redevelopment, Hazzard claimed that:

> The major plus of all this which is not necessarily given enough focus is that when you get a mixture of housing, private and public, you get an incredible drop in antisocial behaviour […] Also what appeals to me is that little kids growing up don't just see their parents and grandparents not going to work […] They see people around them dressed for work, leaving home to go about their day, and they think 'Gee whiz that's what I can do'.[44]

It is important to note that social housing in Australia is a highly residualized tenure. The small stock and tight eligibility criteria result in an increasing number of allocations to people with complex needs. It is the combination of eligibility criteria, limited supply, and spatial distribution that leads to concentrated disadvantage.[45] Yet government policy has focused only on the latter, with deconcentration via mixed tenure (re)development prevailing as the 'common-sense solution' in spite of thin evidence to support claims that it reduces tenants' disadvantage at all, let alone more than alternative or complementary interventions such as higher levels of income support

[40]Jacob Saulwick, "Revealed".
[41]Committee for Sydney, *Planning for Growth*; NSW Productivity Commission, *Building More Homes*.
[42]Rogers and Gibson, "Unsolicited Urbanism".
[43]Heathcott, "Pruitt-Igoe," 450–1; also see Darcy, "From High-rise Projects to Suburban Estates"; and Bridge et al. (eds.), *Mixed Communities*.
[44]Saulwick, "Revealed".
[45]Sisson, "Public Housing and Territorial Stigma".

and social services.[46] Furthermore, in limiting the analysis of 'mix' to the boundaries of an estate, the discourse of deconcentration ignores neighbourhood-scale tenure mixes – the more relevant scale according to several housing scholars.[47] In the case of Waterloo, neighbourhood-scale mix is already approximately 30% social and 70% private housing.

Despite its subordination to other objectives, there is a stated desire among politicians and public servants to improve the quality and suitability of housing for social housing tenants in Waterloo. However, there appears to be some discordance between the statements of politicians and news media on the one hand and those of planning practitioners on the other. Recent planning documents claim that the estate's buildings 'have high maintenance costs and are at the end of their economic life'.[48] They also note that more than half of the estate's dwellings are two-bedroom apartments whereas two-thirds of households are single persons – problematized as a 'mismatch' in the context of a highly targeted and fiscally-constrained public housing system.[49] They describe the estate as 'an "island" in nature with roads physically disconnected, an a-typical block structure and built form, and typologies that are distinctly different from the surrounding areas',[50] '[a]bove ground car parking [leaving] streets devoid of social activity'.[51] Yet none of these claims speak to the qualities of tenants' actual dwellings, which Hazzard described as 'terrible'.[52]

The nearest thing to a publicly available assessment of the condition of Waterloo's housing stock was conducted by the now-disestablished Redfern Waterloo Authority – the planning agency created by the NSW Government in 2004 to lead renewal throughout the neighbourhood, including the Waterloo and Redfern estates. The *Draft Redfern-Waterloo Built Environment Plan Stage 2* created a redevelopment blueprint that was soon abandoned but which set the stage for the next two decades of state government planning for the estate. It claimed that much of the public housing in Waterloo and Redfern is 'not well suited to residents' needs and nearing the end of its economic life'.[53] It identified issues with outdated facilities such as shared laundries and 'layout, orientation, footprints and design [that] are inefficient and do not address the street', leading to 'limited surveillance of the public domain' as well as 'irregular siting' that caused sub-optimal access to sunlight and ventilation and a lack of well-defined separation of public and private spaces.[54] However, such criticisms were largely reserved for the estate's walk-up apartment complexes, particularly as their lower density made their use of space 'inefficient'; relocation of sitting tenants would also be more manageable and a greater uplift in density achieved. It is these buildings which will be the focus of the first stage of the estate redevelopment, along with the mid-rise Dobell, Drysdale, Madden Place and Camelia Grove buildings. Together they comprise 'Waterloo South', the precinct designated in 2020 to progress as a priority, before finalizing plans for the parts of the estate occupied by the higher-density towers and slab buildings.

While the Redfern Waterloo Authority took issue with the 'unsightly … blank, unfriendly' car parks surrounding the towers and slabs, it suggested that the 'economic life of the high-rise buildings (8 or more storeys) are [*sic*] estimated to be a further 30 years', and that, as such, '[i]nfill

[46]Arthurson, *Social Mix and the City*; Bridge et al. (eds.), *Mixed Communities*.
[47]Le Galès "Social Mix and Urban Policy"; Capp et al., "Re-scaling Social Mix"; Darcy and Rogers, *Finding the Right Mix*.
[48]NSW Department of Planning, Industry & Environment "Waterloo South Planning Proposal," 23.
[49]Ibid., 26.
[50]Turner Studio, "Waterloo Estate Waterloo South".
[51]City of Sydney, "Urban Design Study Part 2".
[52]Hazzard, "AHURI Conference Opening Address".
[53]Redfern-Waterloo Authority, *Draft Redfern-Waterloo Built Environment Plan Stage 2,* 3.
[54]Ibid., 33, 58.

development will need to respond to these buildings'.[55] A response from the City of Sydney largely concurred with this assessment, while also making the recommendation that the Dobell and Drysdale buildings 'be retained and refurbished to save money, conserve resources and to retain a valuable record of examples of better designed public housing from the 1980s to be found anywhere in the world'.[56]

Ironically, it is the high rises which attract the most negative attention in popular media, their stigmatization helping to legitimize redevelopment. This is despite the fact that redeveloping them would yield quite limited benefits in the terms articulated by the state government's planning analysis, and have intense social impacts on existing residents.[57] The estate has been described as a series of 'grey concrete buildings … strikingly incongruous against the relatively flat, terraced terrain of their urban surrounds',[58] or more floridly as 'Soviet-style',[59] 'bleak, grey housing commission towers'.[60] Like other high-rise public housing complexes, the towers are occasionally labelled 'Suicide Towers'.[61] The towers and slabs are routinely used as stock images in any story about public housing or housing affordability. Hazzard's successor as Minister for Social Housing, Pru Goward (who was also his predecessor), described these buildings and those in the neighbouring Redfern estate as 'honeypots' for illicit drug dealing and use when announcing a Local Allocation Policy that would prevent any applicant with a drug conviction from moving into high-density public housing in Waterloo, Redfern, and Surry Hills.[62] When the project's proponents discuss the need for new social housing to be 'tenure blind', it is these buildings that they invoke.

As such, the Waterloo estate is subject to multiple forms of stigmatization. Its stigmatization in housing and social policy discourses centres on concentrated disadvantage and the problems that purportedly arise. Its stigmatization in urban planning and design discourse centres on the outmoded characteristics of the earlier housing stock in the south of the estate, including irregular and inefficient siting and ambiguous boundaries of public and private space. It is stigmatization in the media centres on the newer, more visible buildings – the towers and slabs – which are seen as an affront to popular taste and serve to anchor the social disrepute of public housing tenants. The relationship between these discourses and housing quality is tenuous at best: none speak directly to the qualities of tenants' homes. Indeed, stigmatization allows proponents of redevelopment to elide the question of existing housing quality. These discourses open up a powerful 'reputational gap',[63] devaluing the present condition of the estate in relation to what it might in the future become, and thus help legitimize the proposed redevelopment.

## Conclusion

Is the Waterloo estate's housing stock among the worst quality and most in need of replacement out of all public housing in NSW? While it is impossible to answer definitively, there is no evidence to suggest that it is and it is certainly doubtful that it is the case. There are public housing complexes that are older than Waterloo, and several that exhibit just as many if not more of the

[55]Ibid., 4, 54.
[56]City of Sydney, "Submission to Redfern-Waterloo".
[57]Wynne and Rogers, "Emplaced Displacement".
[58]Visenten, "Waterloo Public Housing Tenants".
[59]Visentin "Hundreds of Elderly Residents".
[60]Carswell "Hope Returns".
[61]Carswell, "Suicide Towers".
[62]"Bizarre Plan".
[63]Kallin, "Opening the Reputational Gap".

design shortcomings with which it has been charged. Yet these are not currently slated for redevelopment; as we have discussed, housing quality is just one reason among several driving the redevelopment of Waterloo, and far from the most significant. The estate is a large plot of high value and publicly owned land, with a significant gap between ground rents realized by the state and ground rents potentially received through partial privatization and redevelopment.[64] It also has a significant reputational gap, created through denigration of its present use and idealization of its future use.

Waterloo is therefore *selectively represented* as outdated and unfit for purpose, in the sense of a disproportionate focus on the Waterloo estate in relation to other public housing estates *and* in the identification and articulation of only its negative characteristics that help justify its demolition. All buildings are tarred with the same brush, foreclosing rational assessment of the qualities of the estate's housing stock and consideration of alternatives. As this paper shows, there are many redeeming features of this housing which, alongside the negative impacts of relocation on residents (and negative environmental impacts of demolition and redevelopment), should be weighed against the benefits of replacing it with homes that meet contemporary standards and expectations. A more fulsome evaluation of these benefits and costs suggests that retaining at least some of the estate's existing buildings, renovating, and refurbishing, perhaps developing infill housing and gradually phasing out the established buildings over a longer time period (as well as building or buying new public housing elsewhere) could produce better outcomes. A new rationale will be needed in order for this kind of evaluation to occur – one that prioritizes the wellbeing of existing tenants and maximization of *current* public housing stock in the context of immense immediate and future need.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

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[64]Smith, *The New Urban Frontier*.

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